

From Analog Records to Computational Research Data: Building the AI-Ready Lab Notebook

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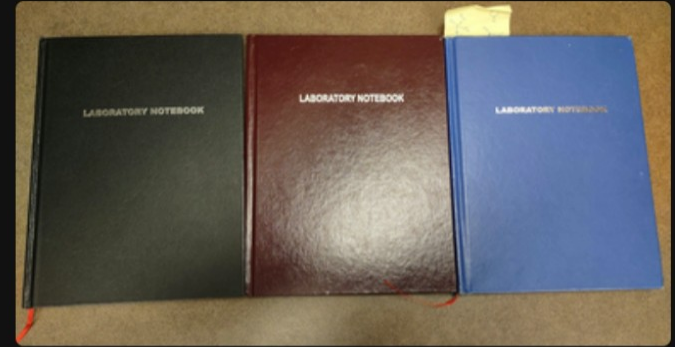
Institute for
Data Driven
Dynamical
Design
ID₄



DREXEL UNIVERSITY
**Metadata
Research Center**
College of Computing & Informatics

Background

- Notebooks collections have potential to provide new insights into the successes, failures and pedagogy of experimental research
- Un-digitized notebooks become "data at risk"
- Research is needed to improve the ease of converting paper lab notebooks into digital, "AI-ready" archival records



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Goals

Overarching Goal: Advance methods for converting scanned handwritten notebooks into structured, FAIR, AI-ready data.

1. **Extraction:** Extract and archive data from notebooks to support computational analysis of these analog artifacts.
2. **Correction:** Identify common error patterns in extraction outputs and build a post-processing pipeline to correct them.
3. **Analysis:** Perform document classification and clustering analysis to answer scientific questions

Methodology

1. Extract contents from pages via optical character recognition (OCR)
2. Post-process output to improve accuracy
3. Build an archive that can be presented by/in the style of an ELN
4. (Work in Progress) Further bolster the AI-readiness of the data by tokenizing protocols

PROJECT 2- Aldehyde protection

Handwritten notes on a grid background:

Eq	FLV	mmol	g/mol	d	mL	Reagent
1.6	235.08	2.127	0.500	-	-	-
1.3	62.07	2.165	0.172	1.1	0.165	Ethylene glycol
0.02	170.22	0.043	0.008	-	-	p-Toluenesulfonic acid
1.0	279.13	2.127	0.544	-	21.3	Toluene

- A clean dry 100 mL RBF was fitted w/ a stirrer.
- Q was added followed by toluene, then Ethylene glycol, then acid.
- Dean-Stark trap was set up.
- Reaction lowered into aluminum bead bath ~140°C.
- After 24 hours Dean-Stark trap was checked, working correctly, so solution moved to heating mantle @ 120°C w/ aluminum foil.
- Heated to 300°C to collect H₂O.
- After 45 min @ 300°C temp lowered to 140°C since H₂O had been removed.
- Reaction quenched after 12 hours at 140°C with NH₄CO₂.
- Extracted with EtOAc but H₂O was added due to salt crashing out upon organic addition to the aqueous phase.

TL: 10/1/2020
Signature: [Signature]
Date: 7/1/16



- A clean 100 mL dry round-bottom flask (RBF) was fitted with a stirrer.
- Q was added, followed by toluene, then ethylene glycol, and then acid.
- A Dean-Stark apparatus was set up.
- The reaction was lowered into an aluminum bead bath at approximately 140°C.
- After 24 hours, the Dean-Stark trap was checked; it was working correctly, so the solution was moved to a heating mantle at 120°C with aluminum foil.
- Heated to 300°C to collect H₂O.
- After 45 minutes at 300°C, the temperature was lowered to 140°C since H₂O had been removed.
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EQ	FW	mmol	g	D	ML	Reagents
1	325.74	.06/40	.02	-	-	HHTP
1.7	237.71	.1044	.025	-	-	tetrafluoro diboronic acid.
						Product
1	-	.06140	.04	-	-	COF-5-F ₄
					2.0	1:1 DIE/Dioxone .03M



```
,EQ,FW,mmol,/2,D/,ML,Reagents
0,1.0,325.74,.06/40,0.02,-,-,HHTP
1,1.7,237.71,.1044,".025",-,-,tetrafluoro diboronic acid.
2,,,/ /,,,Product
3,1.0,,, .06140,".04",-,-,COF-5-F
4,,,,".04",,2,1:1 DIE/Dioxone .03M
```



Reagents	EQ	FW	Mmol	G	D	ML
HHTP	1	325.74	0.0614	0.02	-	-
tetraflu...	1.7	237.71	0.1044	0.025	-	-
Product	-	-	-	-	-	-
COF-5-F	1.0	-	.06140	.04	-	-
1:1 DIE/...	-	-	-	.04	-	2

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Pages

demetrius1-001
demetrius1-003
demetrius1-005
demetrius1-006
demetrius1-007
demetrius1-008
demetrius1-009
demetrius1-011
demetrius1-012
demetrius1-013
demetrius1-014
demetrius1-016
demetrius1-019
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demetrius1-027
demetrius1-028
demetrius1-030

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demetrius1-019
demetrius1-020
demetrius1-021
demetrius1-022
demetrius1-023

Predictions Table Text

Handbook No. _____
Experiment No. _____

DATE: 5/18/15

REAGENT: 1. HHTP 2. 1,4-phenylene diisocyanate

COF: 5

EQ	FW	mmol	g	D	ML	Reagents
1	325.74	.921	.300	-	-	HHTP
1.5	165.78	1.38	.229	-	-	1,4-phenylene diisocyanate
			.500	-	-	COF-5

PROCEDURE: 1. HHTP in DMSO (1.0 g) + 1.0 g of 1,4-phenylene diisocyanate. 2. In Oven at 12-14°C for 5/18/15. 3. Took out of oven at 5/18/15 at 2.00 pm. 4. Filtered with HHTP and used to effect 9.5 g. 5. To get rid of extra impurities. 6. Obtained 3.68 mg of Product and then immediately stored in vial. 7. 3.15 pm. 8. Skipped when necessary. 9. 3.15 pm. 10. Took out of oven, removed the HHTP and the compound and the product was very clean and very pure. 11. 3.15 pm. 12. Vacuum and stored in vial. 13. 3.15 pm. 14. 3.15 pm. 15. 3.15 pm.

Post submitted by: _____
Reviewed by: _____

Predictions Table Text

EQ	FW	mmol	g	D	ML	Reagents
1	325.74	.921	.300	-	-	HHTP
1.5	165.78	1.38	.229	-	-	1,4-phenylene diisocyanate
			.500	-	-	COF-5

Reagents	EQ	FW	Mmol	G	D	ML
HHTP	1	325.74	.921	0.3	-	-
1,4-phenylene...	1.5	165.75	1.38	0.229	-	-
-	-	-	-	-	-	-
COF-5	-	-	-	0.529	-	-

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- A clean dry 100 mL RBF was fitted w/ a stir bar
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- Dean Stark trap was set up
- Reaction lowered into aluminum Bead bath ~140°C
- After 24 Dean Stark trap wasn't working right so solution moved to heating Mantle @ 120°C w/ Al foil
- Heated to 300°C to collect H₂O
- After 45 min @ 300°C temp lowered to 140°C since H₂O had been removed
- Rxn quenched after 12h @ 140°C w/ NH₄CO₃
- Extracted w/ EtOAc but H₂O was added due to salt crashing out upon organic addition to aqueous phase

↓ OCR

- A clean 100 mL dry round-bottom flask (RBF) was fitted with a stirrer.
- I was added, followed by toluene, then ethylene glycol, and then acid.
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- The reaction was lowered into an aluminum bead bath at approximately 140°C.
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- Heated to 300°C to collect H₂O.
- After 45 minutes at 300°C, the temperature was lowered to 140°C since H₂O had been removed.
- The reaction was quenched after 12 hours at 140°C with NH₄CO₃.
- Extracted with EtOAc, but H₂O was added due to salt crashing out upon organic addition to the aqueous phase.

↓ Structure Contents

Action	Entity	Duration	Temperature	Size
dried	RBF			100 mL
fitted	stirrer			
added	I			
added	toluene			
added	ethylene glycol			
added	acid			
set up	Dean-Stark apparatus			
lowered	aluminum bead bath		140C	
checked	Dean-Stark trap	24 hours		
moved	heating mantle		120C	
heated			300C	
lowered		45 minutes	140C	
quenched	NHCO ₂	12 hours		
extracted	EtOAc			

Content Extraction

Recently, the capabilities and performance of document content extraction software have rapidly increased



Content Extraction

Two step Detectron2 and HandwritingOCR method

olmOCR 2

Under N_2 gas

Triazole Condensation

Reaction 700

Reagents: CH_3 , NH_3 , CH_3COOH , Acetic Acid

Br ϕ ϕ ϕ (6)

EQ	FW	mmol	gram	d	AL	Reagent
1.0	266.01	0.683	0.250	-	-	Benzaldehyde
1.0	106.12	0.683	0.072	1.04	0.07	Aniline
1.0	93.13	3.415	0.318	1.02	0.312	Ammonium Acetate
1.0	77.08	2.732	0.211	-	5.46	Acetic Acid 0.125M
-	22.25	0.683	0.361	-	-	(6)

Next 600

A clean dry 15 ML 2NF was fitted w/ condenser

NH_4OAc was added & flame dried

O was added & system was purged w/ N_2

Acetic Acid was added followed by PhOH & PhNH₂

System heated to reflux

After 1 hour solution turned from orange to grey

After 3 hours rxn cooled to room temp & quenched

W/ the H_2O was filtered off & washed w/ the H_2O

Residue was green paste & taken up in DCM to remove the H_2O w/ Br.Ne

had a 50% DCM in HPLC column 25% to 50% to 75% to 95% in product

Yield = 182.8 mg

50.6%

HLMR = $\checkmark$ (CDCl₃)

TLC

100% DCM

8/19/2016

Continued on Page

Read and Understood By

Signed

Date

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490 tokens processed [Copy](#)

3/11/15

DAV-1-7

DOA + HHTP $\rightarrow$ COF-S

EQ	MW	mmol	g	D mL	Reagents
1	325.74	0.0368	0.12	-	HHTP
1.5	165.75	0.543	0.09	-	DOA
				.5	Dioxane
				.5	Mesitylene

5-tubes. In Oven @ 3:12 pm 3/11/15

A - Not COF-S after px11 B - COF-S after px11 C - Not COF-S * D - COF-S * E - Spilled vial (no product)

Obtained 17 mg of Product Then immediately started Solvent exchange.

.005g after activation. 3/11/15

.0037g of DAV-1-6-C

.0035g of DAV-1-6-B 12.2mg of COF-S in glove box

Post Processing

- Regardless of the extraction method, there are always some errors.
- Checks and fixes for tables in RSM DL notebooks include:
 - Removing any empty rows/columns and realigning elements.
 - Correcting header errors (e.g., reading "FW" as "MW").
 - Validating that numerical columns contain only numbers.
 - Correctly handling slashed/blacked out cells
 - Within rows: $FW * mmol = \text{grams}$
 - Between columns: $mmol_i / Eq_i = mmol_j / Eq_j$ for rows i and j .

EQ	FW	mmol	g	D	mL	Reagents
1	325.74	.06140	.02	-	-	HHTP
1.7	237.71	.1044	.025	-	-	tetrafluoro diboronic acid.
						Product
1	-	.06140	.04	-	-	COF-5-F ₄
					2.0	1:1 DIE/Dioxone .03M



```
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2,,,/ /,,,Product
3,1.0,,,06140,".04",-,-,COF-5-F
4,,,,".04",,2,1:1 DIE/Dioxone .03M
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Reagents	EQ	FW	Mmol	G	D	ML
HHTP	1	325.74	0.0614	0.02	-	-
tetraflu...	1.7	237.71	0.1044	0.025	-	-
Product	-	-	-	-	-	-
COF-5-F	1.0	-	.06140	.04	-	-
1:1 DIE/...	-	-	-	.04	-	2

Archival and Presentation

- Following extraction and post-processing, an archive of the contents is generated
- Archive can be rendered in ELN-like web interface, or potentially imported into full fledged ELN in the future

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demetrius1-022
demetrius1-023

Predictions

Table Text

Handbook File: DAV-1-23
Demetrius1-001

5/18/05
EQ = HATP
1.5 = 1.4-phenyl...
100% = 100%
100% = 100%

EQ	FW	mmol	g	D	ML	Reagents
1	325.74	.921	.300	-	-	HATP
1.5	165.75	1.38	.229	-	-	1.4-phenyl...boronic acid
			.529	-	-	COF-5

Notes:
- Followed Procedure in DAV-1-22
- Only difference was a 5mL 1:1 solution
- In Oven at 120°C for 5/18/05
- Took out of oven at 5/22/05 at 2:00pm
- Filled with THF and used to about 9 times
- To get rid of extra impurities
- Obtained 3.68mg of product and then
- immediately shifted solvent exchange to 1:1 THF
- 1.4-phenyl...boronic acid
- Took out of oven at 5/22/05 at 2:00pm
- and the product was very clean and very pure (COF-5)
- Vacuumed and stored in Erlenmeyer flask
- 2 May 05 40 in 2014

Revised on Page: _____
Revised by: _____

Predictions

Table Text

EQ	FW	mmol	g	D	ML	Reagents
1	325.74	.921	.300	-	-	HATP
1.5	165.75	1.38	.229	-	-	1.4-phenyl...boronic acid
			.529	-	-	COF-5

Reagents	EQ	FW	Mmol	G	D	ML
HATP	1	325.74	.921	0.3	-	-
1.4-phenyle...	1.5	165.75	1.38	0.229	-	-
-	-	-	-	-	-	-
COF-5	-	-	-	0.529	-	-

Results

Technique	Mistakes	Error Rate	Improvement Rate
D2+HOCR*	199	18.22%	-
olmOCR 2	61	5.59%	-
D2+HOCR Post-Processed	62	4.76%	73.87%
olmOCR 2 Post-Processed	46	4.21%	24.69%

30 page sampling contains **1,092** cells

*Detectron2 and HandwritingOCR

Future Work

- Now working on converting the contents of text blocks into a tokenized list of actions
- Work is an extension of previous work by our group using LLMs to tokenize methodology sections from MOF research papers (4)
- Tokenizing protocols can greatly improve ML analyses of their content (5), increasing data's "AI-readiness" and the potential for knowledge discovery

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- Extracted w/ EtOAc but H₂O was added due to salt crashing out upon organic addition to aqueous phase

↓ OCR

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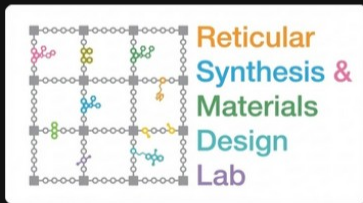
Action	Entity	Duration	Temperature	Size
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added	I			
added	toluene			
added	ethylene glycol			
added	acid			
set up	Dean-Stark apparatus			
lowered	aluminum bead bath		140C	
checked	Dean-Stark trap	24 hours		
moved	heating mantle		120C	
heated			300C	
lowered		45 minutes	140C	
quenched	NHCO2	12 hours		
extracted	EtOAc			

Conclusion

- Overall Goals:
 - Make data contained in lab notebooks AI-ready
 - Mine information in notebooks for scientific insights
 - Content extraction aspects of work largely completed
 - Work to tokenize and further process digital notebook archives to bolster AI-readiness ongoing, moving towards using notebooks to answer scientific questions
- 1. Cliff Pickover [@pickover]. Physics. Marie Curie's (1867–1934) century-old experimental notebook [Internet]. Twitter. 2022 [cited 2025 Dec 4]. Available from: <https://x.com/pickover/status/1526948489099464704> (2)
 - 2. STARLIMS. Electronic Laboratory Notebook (ELN) Software - STARLIMS [Internet]. 2023 [cited 2025 Dec 4]. Available from: <https://www.starlims.com/electronic-lab-notebook/> (2)
 - 3. Zhang J, Liu Y, Wu Z, Pang G, Ye Z, Zhong Y, et al. MonkeyOCR v1.5 Technical Report: Unlocking Robust Document Parsing for Complex Patterns [Internet]. arXiv.org. 2025 [cited 2025 Nov 23]. Available from: <https://arxiv.org/abs/2511.10390v2> (7)
 - 4. Zhao X, Rojas JFF, Furst J, Ardila K, Langlois K, An Y, et al. Expert-Guided LLM Approach for Sequence-Aware Extraction of MOF Synthesis [Internet]. ChemRxiv; 2025 [cited 2025 Nov 23]. Available from: <https://chemrxiv.org/engage/chemrxiv/article-details/689c179c728bf9025e39cb58> (12)
 - 5. Li A, Wang X, Wang W, Zhang A, Li B. A survey of relation extraction of knowledge graphs. In Springer; 2019. p. 52–66. (12)

Thank You!

Questions?



- 1. Cliff Pickover [@pickover]. Physics. Marie Curie's (1867–1934) century-old experimental notebook [Internet]. Twitter. 2022 [cited 2025 Dec 4]. Available from: <https://x.com/pickover/status/1526948489099464704> (2)
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